Booking Cancellation Prediction — INN Hotels

Machine Learning project · Adewale Adeyinka Falade · Python, scikit-learn, statsmodels

Problem

INN Hotels Group was losing revenue to frequent, often last-minute booking cancellations. The objective was to predict high-risk bookings in advance and inform cancellation and pricing policy.

Approach

- Dataset of 36,275 bookings — lead time, guests, price, market segment, booking history, special requests, meal plan.
- Logistic Regression (statsmodels) and a Pruned Decision Tree.
- Categorical encoding, multicollinearity handling, and iterative variable elimination.
- Evaluated on Accuracy, Precision, Recall, F1, ROC-AUC and confusion matrices, with threshold tuning.

Results

Metric	Train	Test
ROC-AUC	86.2%	85.1%
F1 (at threshold 0.37)	—	70.3%

Key Findings

- **Lead time** is the strongest predictor — longer lead times raise cancellation risk.
- Repeat guests, offline bookings and guests with special requests are **less** likely to cancel.
- Higher room prices, no-meal-plan bookings and prior cancellations **raise** risk.
- Dropping `market_segment_type_Online` resolved multicollinearity and stabilised the model.

Recommendations

- Embed the model in the booking platform to flag high-risk cancellations in real time.
- Use a 0.37 decision threshold for the best precision/recall balance.
- Stricter cancellation terms for high-risk bookings; incentives for early confirmation or special requests.
- Feed insights into CRM and pricing systems and build risk-monitoring dashboards.

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